Set Theory

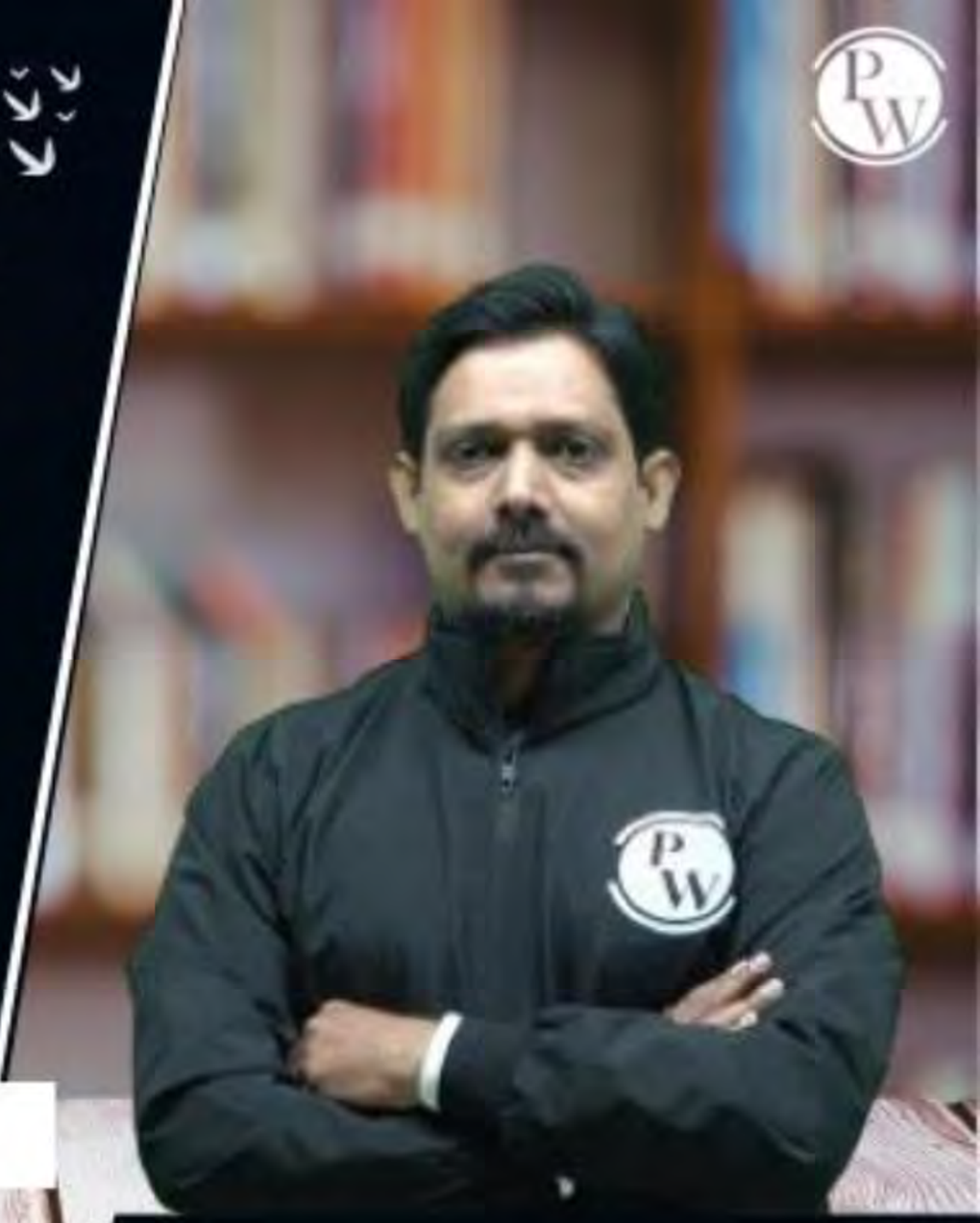
Computer Science & IT



Data Structure

Graph & Hashing

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Counting of Heap : ${}^{n-1}C_k (k) * T(n-1-k)$

↳ BT structure you will find k

Graph :

BFS

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

BFS practice question

Hashing

Slide

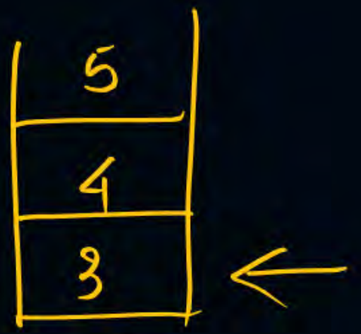
Topic : Graph

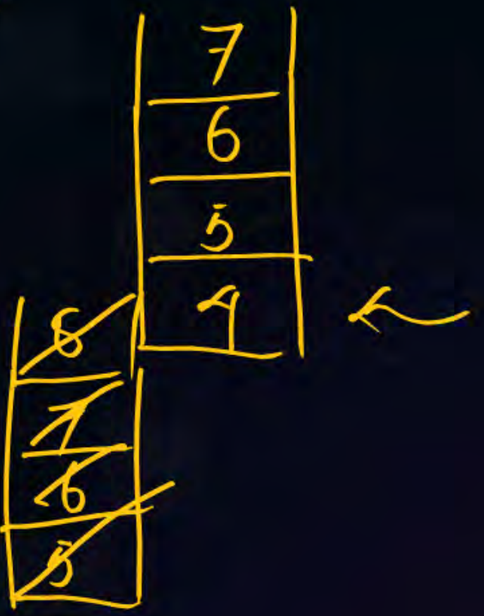


BFS : Data structure Queue









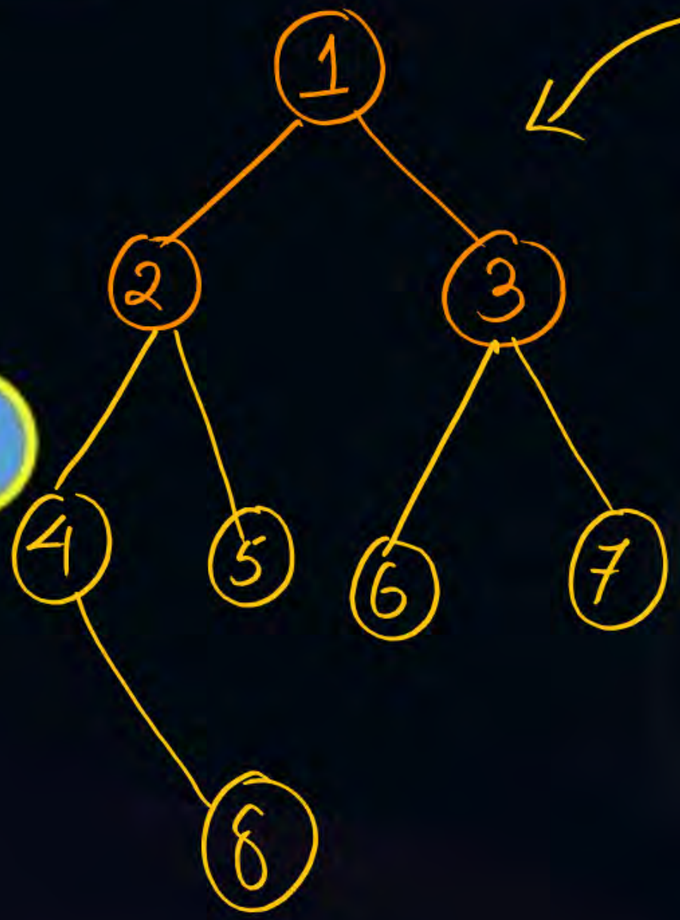
Slide



Topic : Graph



BFS



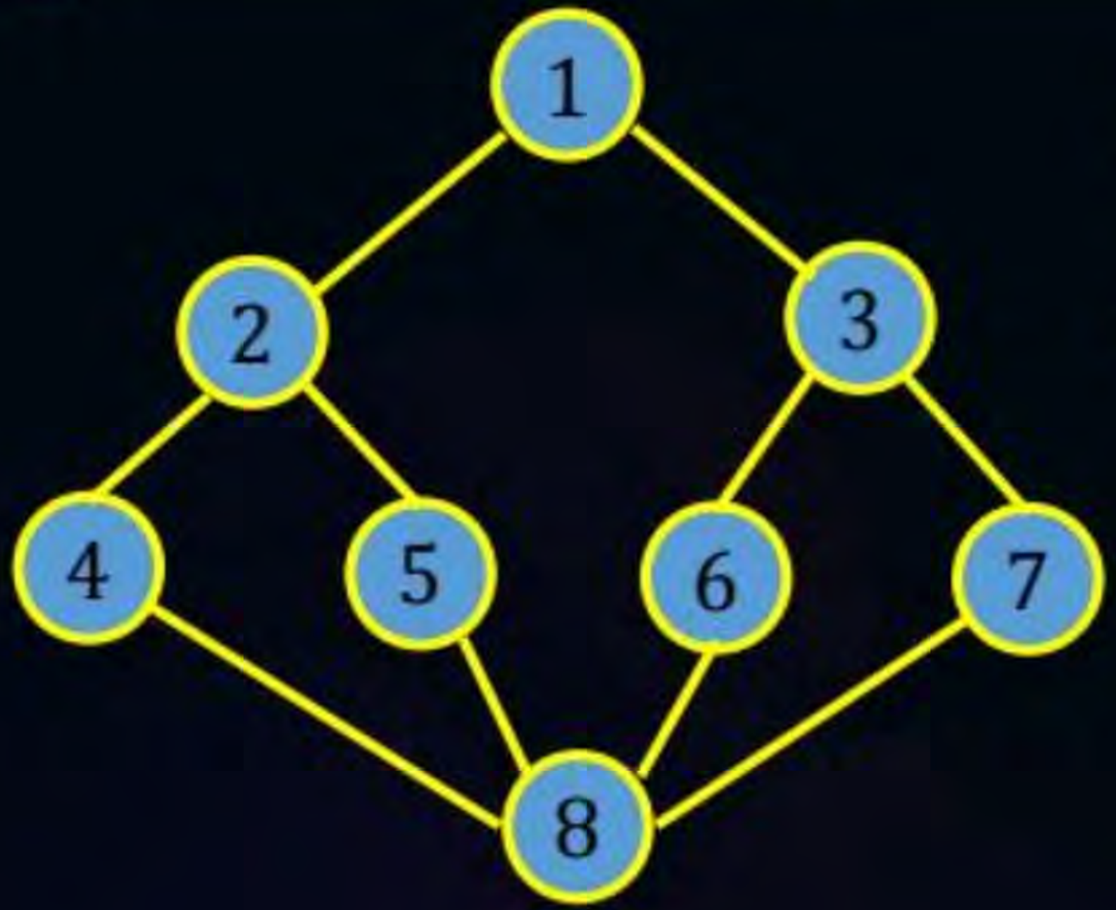
BFS tree

Graph Traversal out come is a Tree called BFS tree

Slide



Topic : Graph

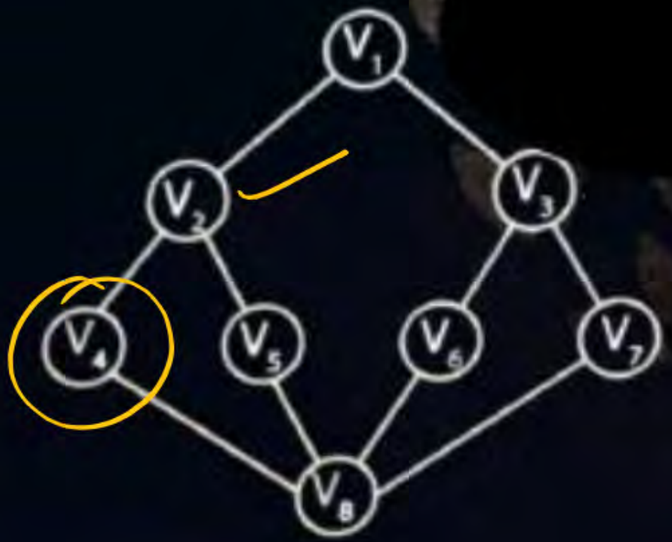


If a Node is already added in BFS traversal it will not be connected by another edge

Slide

Question

Consider the following graph:



Identify **valid BFS** search sequence

(a) 1, 3, 2, 5, 4, 7, 6, 8

(b) 1, 2, 3, 6, 7, 4, 5, 8

(c) 1, 2, 3, 4, 6, 7, 5, 8

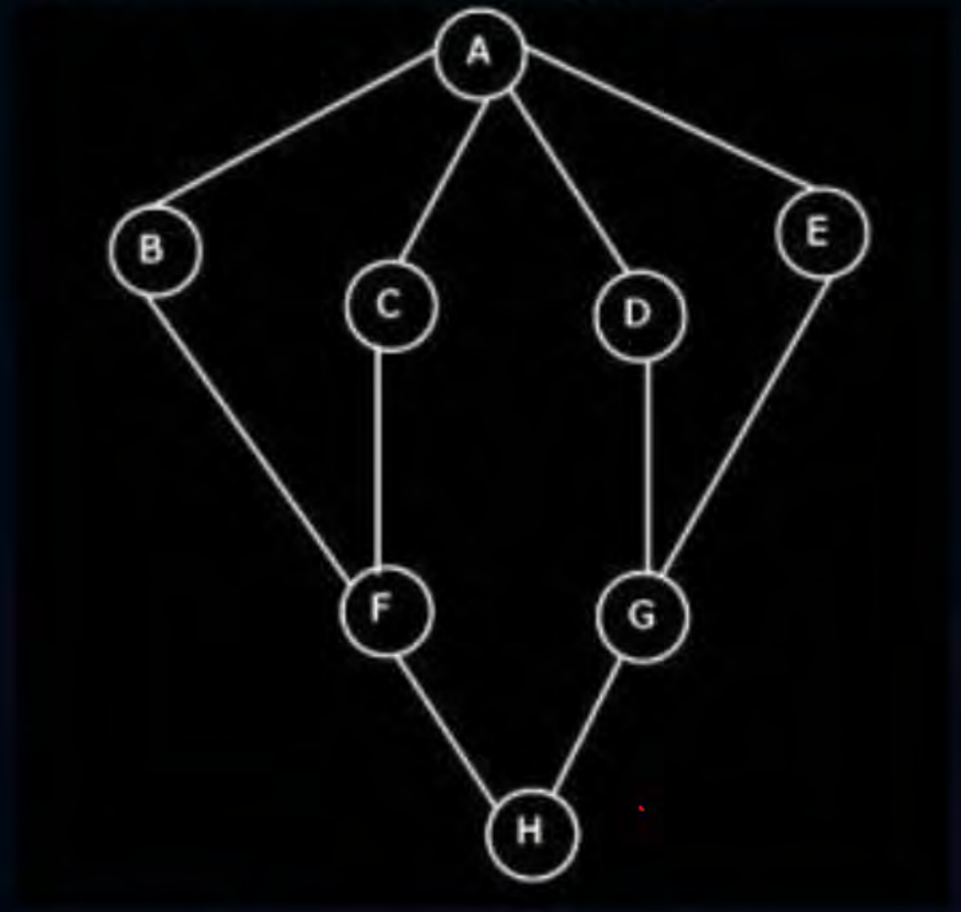
(d) 1, 3, 2, 7, 6, 5, 4, 8

Slide



Topic : Graph

Consider the following graph



How many different breadth-first search traversals are possible considering H as a source vertex?

(A) 1

(B) 4

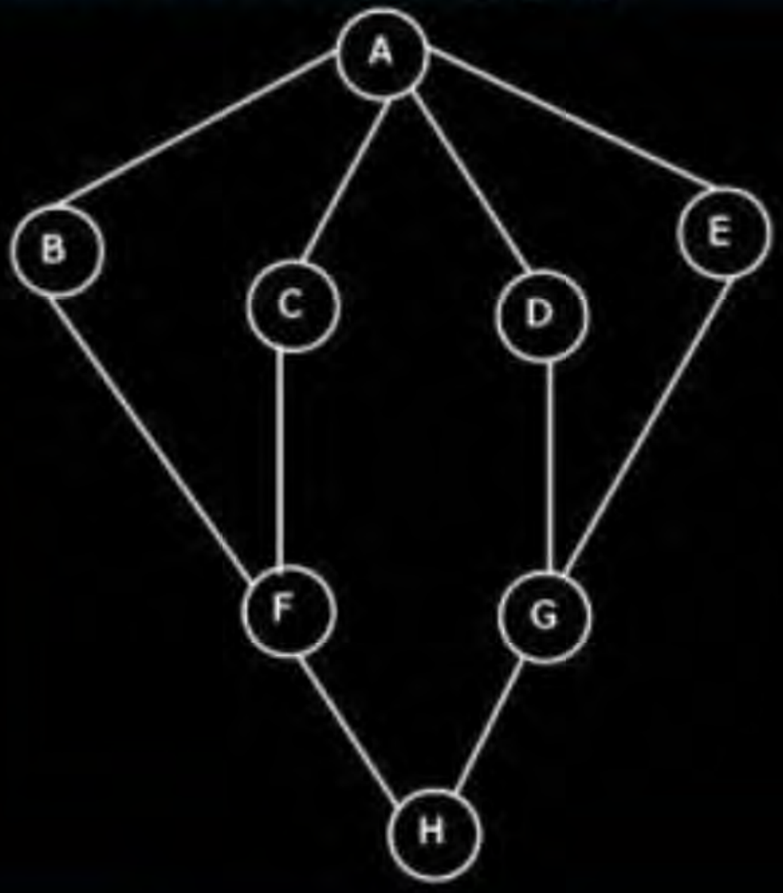
(C) 16

(D) 8

Slide

Topic : Graph

Consider the following graph



H
- F — G
  - B — C
    - D — E — A
    - E — D — A
  - C — B
    - D — E — A
    - E — D — A
- G — F
  - D — E
    - B — C — A
    - C — B — A
  - E — D
    - B — C — A
    - C — B — A

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Slide

Question

Q. Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the nth vertex in this BFS traversal, then the maximum possible value of n is 31.



Slide



Topic : Hashing

Search operation in worst case if n element

* array : Last — Max n comparison
  ordered array — Binary Search - Max $\log_2 n$
* Linked List — Last — Max n comparison

* BST — Leaf Node (Height) — ht+1 comparison - $h = (n-1)$
* AVL — Leaf Node — h+1 comparison
  $h = \log_2 n$

Slide



Topic : Hashing

Direct Addressing : Element is searched by key value

unique attribute

In direct Addressing Index of array represent the key value.

Random Access

Constant time to search the data.

1	D_1
2	D_2
3	D_3
4	D_4
5	D_5
6	D_6
7	D_7

Slide



Topic : Hashing

Direct Address Disadvantage

1. Huge Size of array required

Phone No 9990111019 — a[9990111019]

2. Some Index may Not represent the key value
   Wastage of space.

Slide



Topic : Hashing

Hashing :- Rather than maintaing a big table or array we maintain a Small Size table called Hash table and to distribute the key a function called Hash function is maintained.

Slide



Topic : Hashing

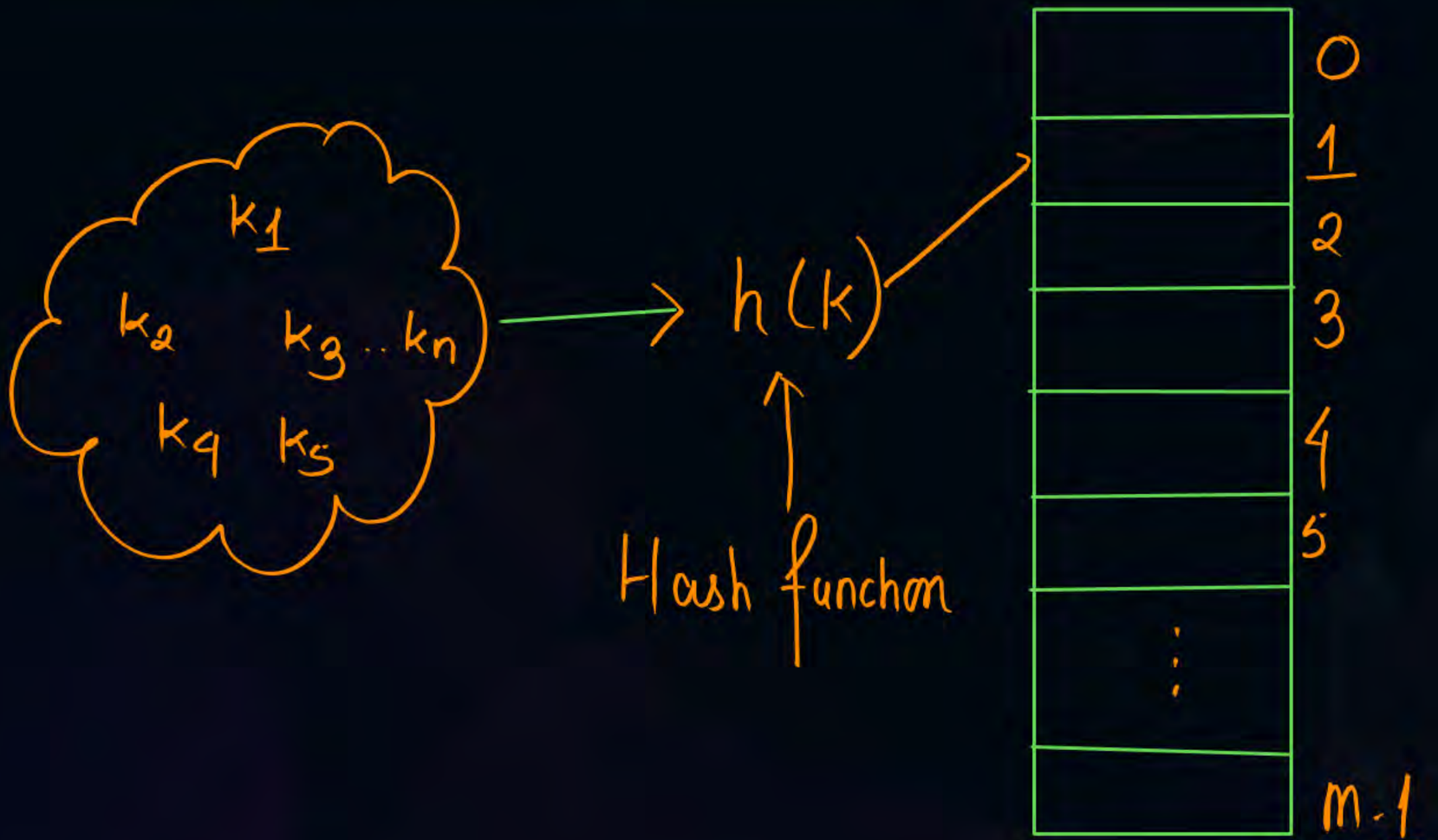
In Direct Addressing The key value k goes to Index k of array

In Hashing The key value k goes to Index h(k)

Slide



Topic : Hashing



Slide



Topic : Hashing

Suppose hash table is of size 10, and hash function is $h(k) = k \bmod 10$



11, 21, 16, 39, 42

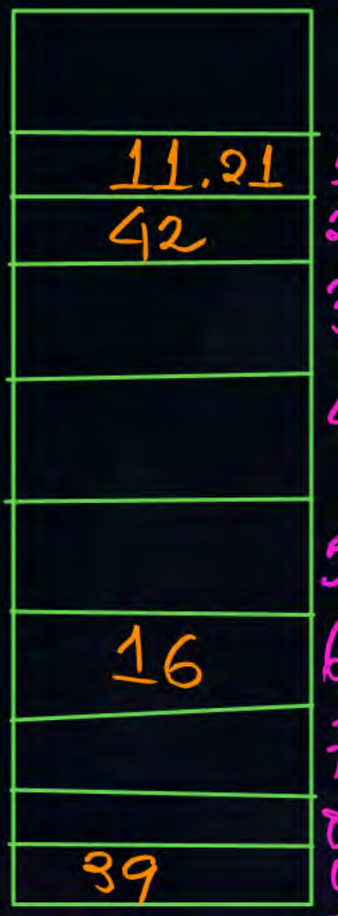
$h(11) = 11\%10 = 1$

$h(21) = 21\%10 = 1$

$h(42) = 42\%10 = 2$

$h(16) = 16\%10 = 6$

$h(39) = 39\%10 = 9$



Two value can be in single place

Slide



Topic : Hashing

1 Hash function should uniformaly distribute the key.

1 choice of Hash function

2. Table Size (m) if a prime No. then it leads to even distribution of keys.

Slide



Topic : Hashing

Which one of the following hash functions on integers will distribute keys most uniformly over 10 buckets numbered 0 to 9?

(A) $h(i) = i^2 \bmod 10$

(B) $h(i) = i^3 \bmod 10$

(C) $h(i) = (11*i^2) \bmod 10$

(D) $h(i) = (12*i^2) \bmod 10$

Topic : Hashing

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

$h(i) = i^2 \bmod 10$

0	10
1	1, 9, 11
2	
3	
4	2, 8, 12
5	5
6	4, 6
7	
8	
9	3, 7

} empty

} empty

$h(i) = i^3 \bmod 10$

0	10
1	1, 11
2	8
3	7
4	4
5	5
6	6
7	3
8	2, 12
9	9

More uniform compare to first one

Bucket / slots

Slide



Topic : Hashing

1. Each location of table is called bucket, slot,
2. The Number of keys to mapped is n

   and Size of table represented as m

Load factor : The ratio between No. of keys to No. of Slots.

$$\alpha = \frac{n}{m}$$

Slide



Topic : Hashing

#Q Given a has table T with 25 slots that stores 2000 elements, the load factor α for T is _____.



Topic : Hashing

#Q Given a hash table T with 25 slots that stores 2000 elements, the load factor α for T is _____.

$$\alpha = \frac{2000}{25} = 80$$

Slide



Topic : Hashing

Collision : Two keys mapped to same location leads to collision

Collision Resolution Technique

1. open Hashing
2. Closed Hashing

Slide



Topic : Hashing

In open Hashing in case of collision collided key store out side of table. one of the Technique called chaining or Seperate chaining forms a linked list for collided keys.

Slide

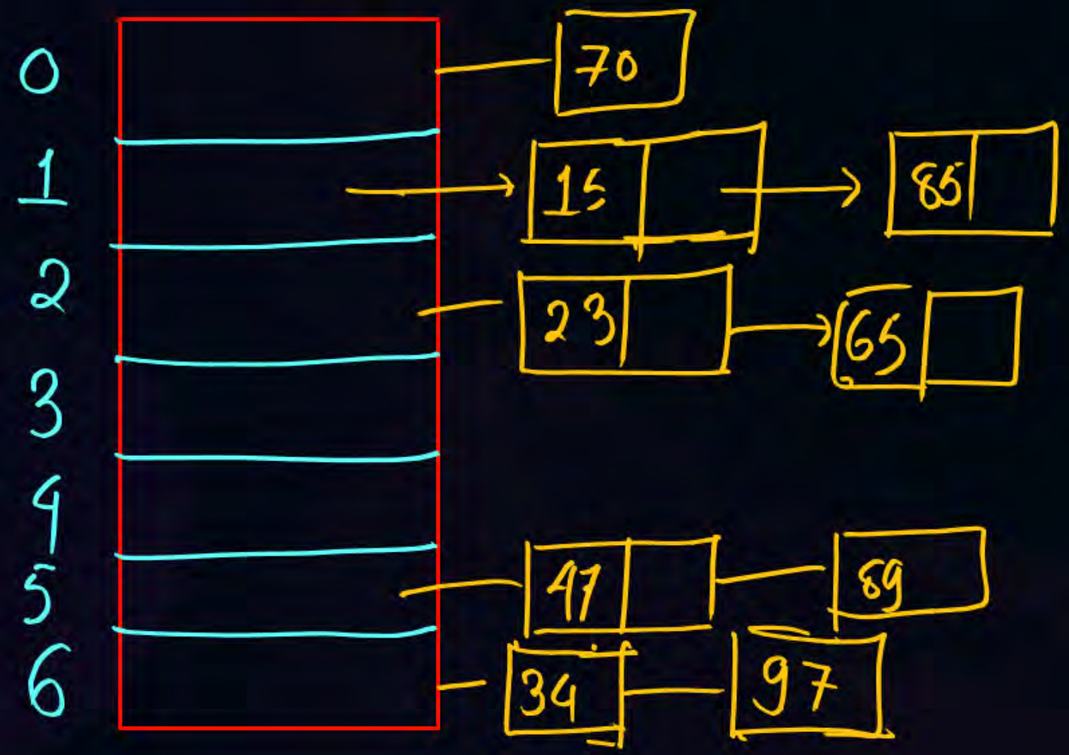


Topic : Hashing

Example

$h(x) = x \bmod 7$

Let us see the following example to get better idea. If we have some elements like {15, 47, 23, 34, 85, 97, 65, 89, 70} . And our hash function is $h(x) = x \bmod 7$.



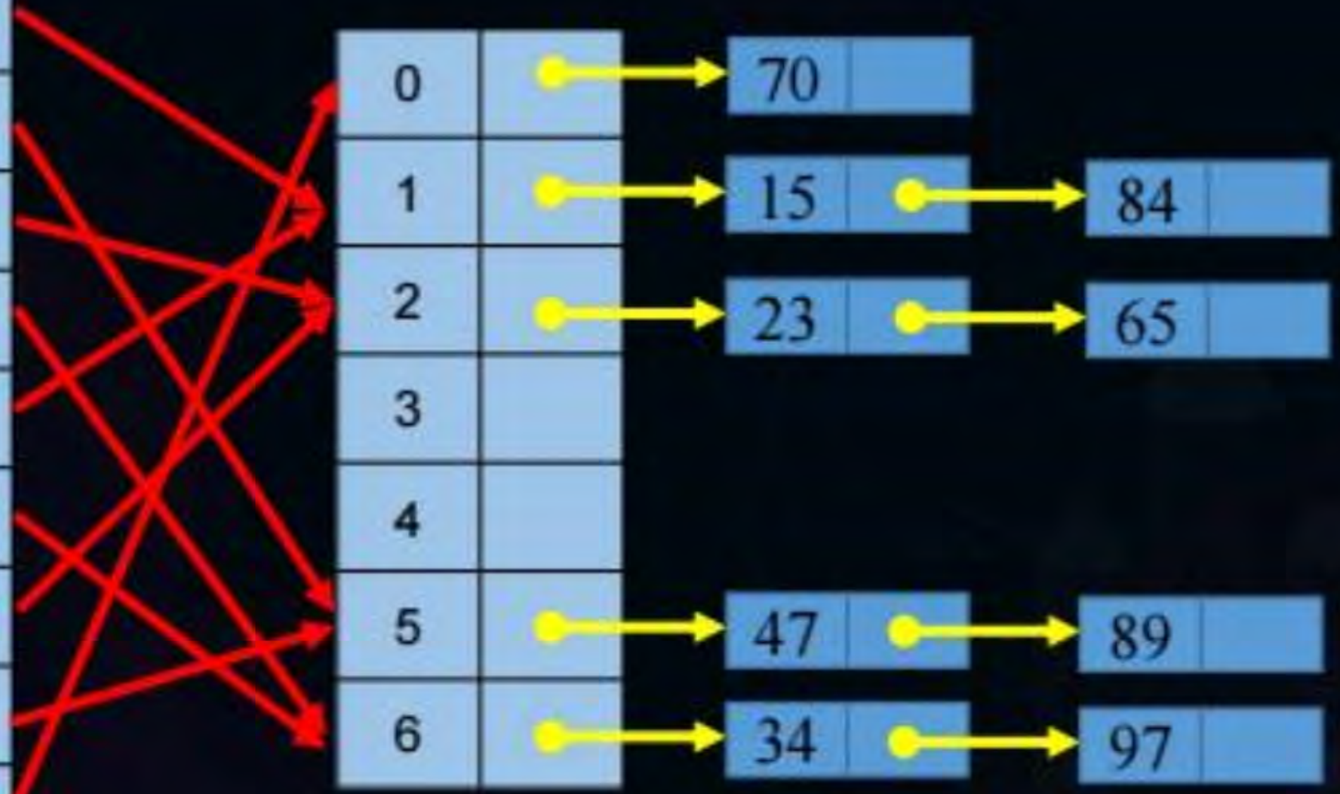
chaining

Slide



Topic : Hashing

$Key - x$	$h(x) = x \bmod 7$
15	1
47	5
23	2
34	6
85	1
97	6
65	2
89	5
70	0



Slide



Topic : Hashing

$h(x) = x \bmod 9$

#Q. Consider a hash table with 9 slots. The hash function is The collisions are resolved by chaining.

The following 9 keys are inserted in the order: 5, 28, 19, 15, 20, 33, 12, 17, 10.

The maximum, minimum, and average chain lengths in the hash table, respectively, are

(A) 3, 0, and 1

(B) 3, 3, and 3

(C) 4, 0, and 1

(D) 3, 0, and 2

Slide

Topic : Hashing

#Q. Consider a hash table with 9 slots. The hash function is The collisions are resolved by chaining.

The following 9 keys are inserted in the order: 5, 28, 19, 15, 20, 33, 12, 17, 10. The maximum, minimum, and average chain lengths in the hash table, respectively, are

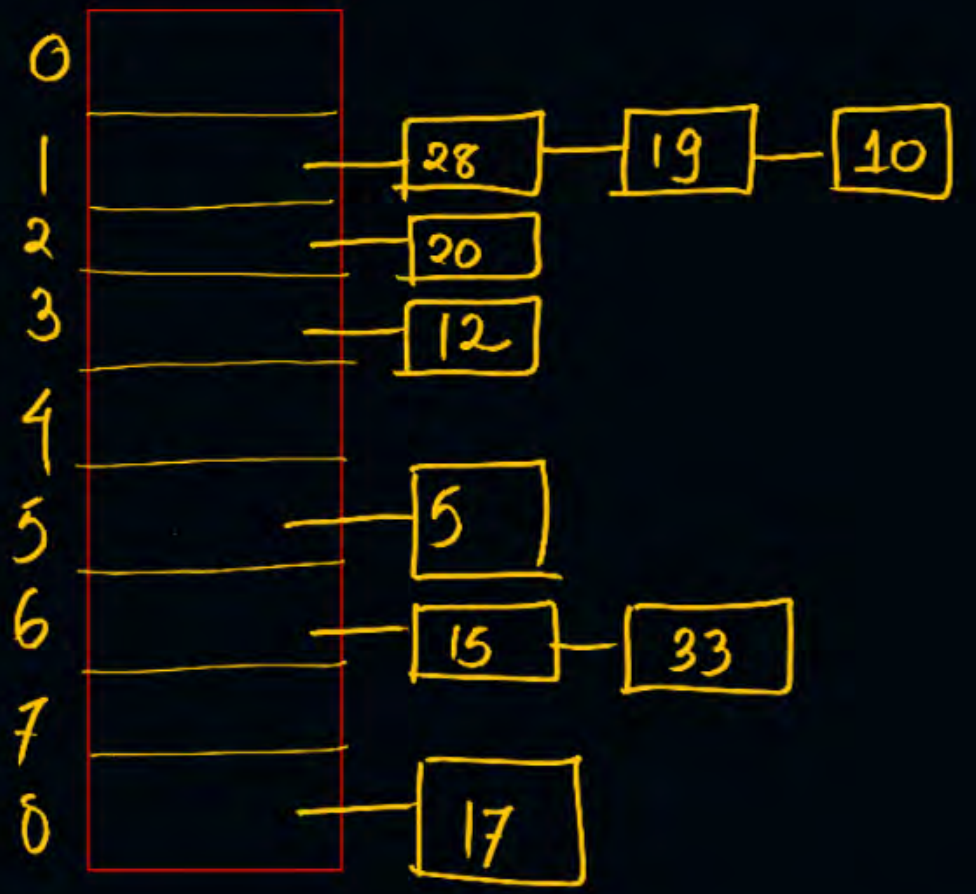
(A) 3, 0, and 1
(B) 3, 3, and 3
(C) 4, 0, and 1
(D) 3, 0, and 2

Slide

Max = 3

Min - 0

Avg = 1





Topic : Hashing

Hw

Q. Consider a hash table with 100 slots. Collisions are resolved using chaining. Assuming simple uniform hashing, what is the probability that the first 3 slots are unfilled after the first insertions?

(A) $(97 \times 97 \times 97)/100^3$

(B) $(99 \times 98 \times 97)/100^3$

(C) $(97 \times 96 \times 95)/100^3$

(D) $(97 \times 96 \times 95)/(3! \times 100^3)$



Topic : Heap

```
Algorithm Adjust(a,i,n){
j ≔ 2i; x:= a[i];
while (j ≤ n) do {
        if ((j < n) and (a[j] < a[j + 1]))
                then j := j + 1;
        if (item ≥ a[j}) then break;
        swap(a[⌊j/2 ⌋]  := a[j]);
        j := 2j;
}

}
```

Slide



Topic : Heap

```
Algorithm DelMax(a,n,x)
{
      if (n = 0) then{
        write ("heap is empty"); return false;
      }
      x := a[1];
      a[1] := a[n];
      Adjust(a, 1, n - 1);
      return true;
}
```



Topic : Heap

```
Algorithm buildheap (a,n){
   for i = ⌊n/2⌋ to  1 step -1 do
   Adjust(a,i,n)
}
```



Topic : Question

#Q. You are given a set V of distinct integers. A binary search tree T is created by inserting all elements of V one by one, starting with an empty tree. The tree T follows the convention that, at each node, all values stored in the left subtree of the node are smaller than the value stored at the node. You are not aware of the sequence in which these values were inserted into T. and you do not have access to T.

Which one of the following statements is TRUE?

Postorder traversal of T can be determined from V

Inorder traversal of T can be determined from V

Root node of T can be determined from V

Preorder traversal of T can be determined from V

Slide



Topic : Question

#Q. You are given a set V of distinct integers. A binary search tree T is created by inserting all elements of V one by one, starting with an empty tree. The tree T follows the convention that, at each node, all values stored in the left subtree of the node are smaller than the value stored at the node. You are not aware of the sequence in which these values were inserted into T. and you do not have access to T.

Which one of the following statements is TRUE?



2 mins Summary

Topic

BFS

Topic

Hashing

Topic

Chaining

Topic

Topic

Slide

THANK - YOU

Slide